Ex. No: 8 Deadlock Detection Algorithm

C PROGRAM

```
#include <stdio.h>
int main()
{
int allocation[10][10], request[10][10];
int available[10];
int work[10];
int finish[10];
int n, m;
int i, j, k;
int found;
printf("Enter Number of Processes: ");
scanf("%d", &n);
printf("Enter Number of Resource Types: ");
scanf("%d", &m);
printf("
\nEnter Allocation Matrix:
\n");

for(i = 0; i < n; i++)
{
for(j = 0; j < m; j++)
{
scanf("%d", &allocation[i][j]);
}
```

```
}
printf("
\nEnter Request Matrix:
\n");

for(i = 0; i < n; i++)
{
for(j = 0; j < m; j++)
{
scanf("%d", &request[i][j]);
}
}
printf("
\nEnter Available Resources:
\n");

for(i = 0; i < m; i++)
{
scanf("%d", &available[i]);
work[i] = available[i];
}
for(i = 0; i < n; i++)
{
finish[i] = 0;
}
do
```

```
{
found = 0;
for(i = 0; i < n; i++)
{
if(finish[i] == 0)
```



```
{
for(j = 0; j < m; j++)
{
if(request[i][j] > work[j])
break;
}
if(j == m)
{
for(k = 0; k < m; k++)
{
work[k] += allocation[i][k];
}
finish[i] = 1;
found = 1;
```

```
}
}
}
} while(found);
found = 0;
printf(
"
\nDeadlocked Processes:
\n");

for(i = 0; i < n; i++)
{
if(finish[i] == 0)
{
printf("P%d ", i);
found = 1;
}
}
if(found == 0)
{
printf("No Deadlock Detected");
}
printf("
\n");
return 0; }
```

Output:

```
┌──(kali㉿kali)-[~]
└─$ ./deadlock
Enter Number of Processes: 3
Enter Number of Resource Types: 3

Enter Allocation Matrix:
1 0 0
0 1 0
0 0 1

Enter Request Matrix:
0 0 0
0 0 0
0 0 0

Enter Available Resources:
1 1 1

Deadlocked Processes:
No Deadlock Detected
```

SHELL SCRIPT

```
#!/bin/bash

echo "Enter number of processes:"

read n

echo "Enter deadlocked process numbers (if any):"

read processes

if [ -z "$processes" ]

then

echo "No Deadlock Detected"

else

echo "Deadlocked Processes:"

echo "$processes"

fi
```

Output:

```
┌──(kali㉿kali)-[~]
└─$ bash lab8.sh
Enter number of processes:
5
Enter deadlocked process numbers (if any):
4
Deadlocked Processes:
4
```